Four pairs in the binary equality case

Human Compression project: a component of the L044 recovery

Let the natural alternating group A_{2m} have a string C-marking of rank m, with $m \geq 8$ even. Let $H = G_i$ be a transitive maximal omission preserving two blocks of size m, called rows, and suppose every distinguished maximal omission H_j is intransitive. Choose a row-swapping mark l of H with two noncommuting neighbours r, s on opposite sides in the string. Thus r, s commute. Let C contain the other $m - 4$ marks.

We acquire the complete contracted-forest theorem. Its graph J consists of all simple contracted edges of r, s on the l-pairs. It has rank three. The graph obtained by adjoining one selected eligible edge for every C label is connected, and every selected C edge is a bridge. A row-preserving R mark is a double transposition with two distinct contracted edges; a row-swapping R mark has a contracted triangle and equals l elsewhere. Contraction is only a graph operation; r, s need not permute the contracted pairs.

Theorem 1. *The nonisolated vertices of J are exactly four pairs, and J is connected on them. If both r, s preserve the two rows, the configuration is impossible.*

The common graph reduction

Write Y for the set of nonisolated vertices of J, and put $K = \langle l, r, s \rangle$. Each component of J lifts to one K-orbit of twice its size. Collapse these components in the graph with the selected C edges; the resulting graph is a tree.

Every $g \in C$ acts on pair indices and has the form $\mathrm{diag}(\pi) l^{\chi}$, with one global row character $\chi \in \{0, 1\}$. Its additional projected edges lie in Y. Indeed a connecting path for an additional edge can contain no selected edge of another C label, since this would bypass that eligible edge. Thus the new edge either lies in one J component, or joins the same two components as its own selected edge. If one of these components is an isolated outside vertex, the matching condition on π excludes a distinct additional edge: that vertex is already used by the selected transposition.

Let c_-, c_+ denote the possible outer C neighbours of r, s. Every other C mark centralizes K. Such a mark preserves each nontrivial J component. Components of sizes three and two give different K-orbit sizes. In the case of three components of size two, their marked actions differ: one has only r, one has both r, s, and one has only s. They cannot be interchanged by an equivariant permutation. Thus only c_-, c_+ can join a nontrivial component to another component.

A mark commuting with l, r preserves the contracted support of r, because this is the union of the $\langle l, r \rangle$-orbits larger than two. The same observation applies to s. An edge to an outside pair must therefore start outside the support of the R mark with which its label commutes.

Lemma 2. *The graph J is connected on four nonisolated vertices.*

Proof. Rank three and the two individual local forms leave component sizes 4, $3+2$, or $2+2+2$ for the nontrivial components. In the last case $|Y| = 6$: each R graph has two disjoint edges, one shared between them. The three nontrivial components are marked-inequivalent as above. Collapsing the outside selected C trees as well leaves at least four vertices to be joined by just the two possible core-joining labels. This is impossible.

Suppose $|Y| = 5$, with components P, Q of sizes three and two. Both R marks preserve rows. If one swapped them, its triangle would fill P and it would equal l on Q. A row-preserving other mark would have just one actual transposition above Q, which cannot commute with l there. Two row-swapping triangles have union rank two on the same support, rank at least four when they meet in at most one vertex, or union on four vertices when they meet in two. None gives the present case.

The outside vertices form one selected tree O: otherwise the collapsed graph has at least four vertices and just two available joining labels. Its two joins, labelled c_-, c_+, connect P, Q, O in a tree. If they are P–O and Q–O, the outside path between them is a noncommutation path. Adjacent projected edges share an outside vertex; additional edges inside Y cannot cancel this noncommutation. Together with the path c_-, r, l, s, c_+, this makes a cycle in the string, impossible.

Otherwise one join g connects P to Q. Suppose g commutes with r, so the other join d commutes with s. The r graph cannot lie entirely in P: it would give a six-point $\langle l, r\rangle$-orbit there and only two-point orbits over Q, prohibiting g. Hence r has one edge on Q and one on two vertices of P. Equivariance forces g to carry the entire four-point orbit over Q onto that over those two vertices of P. Its projected involution exchanges the corresponding two pairs of vertices. It fixes the remaining vertex v of P: its matching has no available partner inside $P \cup Q$, and its only selected edge already joins these components. Connectivity of P in J puts v in the support of s.

If d joins Q to O, its inside endpoint is moved by g. Then g, d do not commute, since g fixes the outside endpoint and d has no other outside edge. If d joins P to O, it cannot leave at v, because it commutes with s. Its endpoint is again moved by g, giving the same contradiction. But the two outer labels c_-, c_+ must commute. Thus $|Y| = 5$ is also excluded, proving the assertion. □

Lemma 3. *After Lemma 2, a C mark whose selected edge lies entirely outside Y centralizes K.*

Proof. Collapse Y to one vertex. The selected edges form a tree; each connected component on outside vertices therefore has exactly one edge to Y, called its exit. Suppose an outside selected label c fails to commute with K. It must be the outer neighbour of r or of s. Follow its outside selected tree towards its exit e. Consecutive edges along this path have distinct labels and fail to commute. If both other endpoints are outside, their projected transpositions share just one outside vertex. If one is inside Y, evaluation at the outside endpoint of the other edge still shows noncommutation. Additional edges inside Y cannot cancel either disagreement.

The exit label also fails to commute with K: a centralizing permutation preserves its unique eight-point orbit and cannot leave it. Thus the path just constructed joins two distinct C labels, each adjacent to an R mark. They cannot both be adjacent to the same R mark, since that mark already has l as its other neighbour. They attach to opposite ends of r, l, s, and their outside path forms a cycle with that core path. The noncommutation graph is a subgraph of a string, so this is impossible. □

Both neighbours preserve rows

Assume this from now on. Two distinct commuting double transpositions have disjoint actual supports, the same four-point support with different Klein matchings, or one common actual transposition with two other mutually disjoint transpositions. Indeed commutation preserves or exchanges their transposition orbits, giving precisely these alternatives.

If their supports coincide with different matchings, all four points lie in one row. A distribution of two points per row admits only one row-preserving matching; odd row sizes cannot support one. Both marks therefore use all four pairs of Y, so neither outer C neighbour can exit. This contradicts transitivity.

Suppose their actual supports are disjoint. If r uses four pairs in one row, the remaining four points are in the other row, so again both supports use all four pairs and no exit exists. If r uses four pairs with one transposition in each row, the only matching on the remaining points is its row mirror; the two contracted graphs coincide, giving rank two instead of three. Finally, if r uses three pairs, name it

$$r = (a_0\ b_0)(a_1\ c_1).$$

The available points force $s = (b_1\ d_1)(c_0\ d_0)$. Direct multiplication gives

$$(rl)^3 = \mathrm{diag}((b\ c))l = (sl)^3.$$

This element is outside $\langle l\rangle$, contradicting $\langle r, l\rangle \cap \langle l, s\rangle = \langle l\rangle$.

In the common-edge case the three simple edges form a path. A star would require three mutually disjoint actual edges at one l-pair, which has only two points. A shared end edge or a shared middle edge gives, up to row interchange, reversal and naming,

$$\begin{aligned} &\mathrm{A}: \quad r = (a_0\ b_0)(c_0\ d_0), \quad s = (b_1\ c_1)(c_0\ d_0),\\ &\mathrm{B}: \quad r = (a_0\ b_0)(a_1\ c_1), \quad s = (a_1\ c_1)(c_0\ d_0). \end{aligned} \tag{1}$$

The actual coloured K graph is an eight-point path. Its edge colours in the two cases are respectively

$$l, r, l, s, l, \{r, s\}, l, \qquad l, r, l, \{r, s\}, l, s, l.$$

Neither path reversal preserves these colours. Hence the centralizer of the K action on these eight points is trivial. By Lemma 3, every selected outside mark fixes the core pointwise. Its global row character is zero and it has no extra edge inside Y. It is precisely the plain doubled selected transposition. The outside selected trees are paths: three incident plain transpositions would give a noncommuting triangle. An exit meets an endpoint of its outside path: two incident tail marks would give it two noncommuting neighbours in addition to its adjacent R mark, impossible in a string.

The exits in the two forms

Lemma 4. *In form* A *there is exactly one exit, at a, and it and every subsequent tail mark are plain doubled transpositions.*

Proof. Only c_- can exit, because r uses all four pairs. Its exit is at a, the only pair outside the contracted support of s. The centralizer of the six-point alternating path for $\langle l, s\rangle$ gives two possibilities, where x is the first outside index:

$$c = \mathrm{diag}((a\ x)), \qquad c = \mathrm{diag}((a\ x)(b\ d))l.$$

There is a tail mark $t = \mathrm{diag}((x\ y))$, since the outside tree has $m - 4 \geq 4$ vertices. In the second possibility,

$$e = (ct)^3 = \mathrm{diag}((b\ d))l = (sl)^3.$$

Thus e lies in both $\langle l, r, s, c\rangle$ and $\langle r, s, c, t\rangle$. It does not belong to $Q = \langle r, s, c\rangle$: the set $\{a_0, b_0, x_1, d_1\}$ is invariant under every generator of Q, but e sends a_0 to a_1 outside it. This contradicts the required intersection being Q. Only the plain exit remains, and Lemma 3 supplies every plain tail mark. □

Lemma 5. *In form* B *there is exactly one plain exit and a plain doubled tail. Up to reversal, the exit is at b.*

Proof. A left exit joins b to an outside index x. Its possibilities, again from the six-point alternating-path centralizer, are

$$c = \mathrm{diag}((b\ x)), \qquad c = \mathrm{diag}((b\ x)(a\ d))l.$$

The right exit is symmetric. In the second possibility put $u = cl = \mathrm{diag}((b\ x)(a\ d))$. Inside $\langle c, r, l\rangle$, the element $(rr^u)^2$ is a nontrivial pure row-one three-cycle on $\{a, c, d\}$. The projection of $\langle r, r^l, u\rangle$ to that row is S_5 on a, b, c, d, x: it contains $(a\ c), (a\ b)$, and their u-conjugates give $(d\ c), (d\ x)$. These transpositions form a connected tree. Conjugating the pure three-cycle therefore gives all three-cycles, hence pure A_5 on that row; conjugation by l gives pure A_5 on the other. Since s differs from r by the pure row-zero double transposition $(a\ b)(c\ d)$, this puts s in $\langle c, r, l\rangle$, contrary to IP.

Both plain exits cannot coexist. Write them as $c = \mathrm{diag}((b\ x))$ and $d' = \mathrm{diag}((d\ y))$. Here $x \neq y$, since otherwise the two exits would not commute. Each of the groups $\langle c, r, l, s\rangle$ and $\langle r, l, s, d'\rangle$ contains $w = \mathrm{diag}((a\ b))$. For the first, rr^c is a pure row-zero three-cycle on $\{a, b, x\}$. The row-zero projection of $\langle r, s, r^l, s^l\rangle$ is S_4 on the four core indices. Its conjugates supply all cycles $(u\ v\ x)$, hence pure A_5; conjugation by l supplies the other row. Multiplying r by a suitable pure row-one three-cycle then gives w. For the second group use $ss^{d'}$ on $\{c, d, y\}$ and the same argument.

However $w \notin K$. On each core row let S_4 act on its three partitions into pairs,

$$ab|cd, \qquad ac|bd, \qquad ad|bc.$$

Let $p, q \in S_3$ be the distinct transpositions induced by $(a\ b), (a\ c)$, and let ϕ be the automorphism exchanging p, q. It satisfies $\phi^2 = 1$, since these transpositions generate S_3. Both r, s induce (p, q) on the two rows, whereas l exchanges rows. The graph of ϕ in $S_3 \times S_3$ is invariant under this exchange because $\phi^2 = 1$. Hence every row-preserving element of K has its two images in that graph. But w has image (p, p), which is outside it. The required intersection of the two end subgroups is K, a contradiction. Exactly one plain exit remains, with its plain tail by Lemma 3. □

The omitted endpoint

In both remaining forms the noncommutations of H give one connected path. Thus the missing parent label is an endpoint; an interior omission would disconnect that path. We exclude both orientations of the missing even involution σ.

In form A, put $O = \{a\}$ together with the outside indices. The C marks generate diagonal S_O. If σ is next to s, it centralizes $\langle l, r, C\rangle$. This group has a large orbit of size $2(m-2)$ over $O \cup \{b\}$, and a four-point orbit over $\{c, d\}$. Products of C-conjugates of r generate pure A_{m-2} on the first row of the large orbit. A centralizing permutation preserves that support and fixes it pointwise, since the centralizer of a natural alternating group of degree at least six is trivial. Transitivity then forces it to fix the whole large orbit. On the remaining four points $\langle l, r\rangle$ is the natural D_8, whose centralizer is $\{1, \mathrm{diag}((c\ d))\}$. Both elements preserve the original binary rows. Hence the missing generator would preserve those blocks, contradicting the natural alternating parent.

If σ is next to the far tail, it centralizes the subgroup with the last tail mark removed. There is one orbit of size $2(m-1)$ and the last isolated l-pair. Put $O' = O \setminus \{\text{last index}\}$. Products of conjugates of r under the remaining C marks give pure A_{m-3} on $O' \cup \{b\}$ in row zero. Here $m - 3 \geq 5$, so the same centralizer argument fixes the whole large orbit. The only remaining nonidentity action is the odd transposition on the last pair, impossible.

In form B, orient the unique exit at b and put $O = \{b\}$ together with the outside indices. At the endpoint next to s, the commuting subgroup $\langle l, r, C\rangle$ has a large orbit of size $2(m-1)$

and the isolated pair over d. Products of C-conjugates of r give pure A_{m-2} on $O \cup \{a\}$ in row zero. Its support and transitivity force the large orbit to be fixed pointwise; only an odd flip of the isolated pair remains. At the other endpoint, use pure A_{m-3} on $O' \cup \{a\}$ instead. Again the large orbit of size $2(m-1)$ is fixed and only the odd flip on the last pair remains. Every orientation is excluded, proving Theorem 1.

Attribution and scope. The historical target and fracture strategy are from Cameron–Fernandes–Leemans–Mixer, Proposition 3.5. The root investigator supplied the common support reduction and the classification, intersection and endpoint arguments in this paper. The continuation investigator prepared this manuscript and added the explicit attachment interface in Lemma 3; this addition requires its own distinct check. The separate component with a row-swapping neighbour uses a new central pairing. No finite diagnostic is a premise, and no whole-source recovery is claimed here.